

The Row Sums Were the Method, Not the Theorem: a Machine-Checked Volume-Free Resolvent Bound, and a Misattributed Uniformity Wall

Lluis Eriksson

Abstract

A formal development of a two-dimensional spatial transfer kernel proves its uniform spectral statements by Schur’s test on *constant* row sums, and carries a theorem, labelled THE OBSTRUCTION in its own source, stating that those row sums stop being constant the moment the spatial coupling is switched on. That theorem obstructs the *method*. It was read as obstructing the *conclusion*, and the reading outlived its own retraction.

We prove, in Lean 4 with no `sorry` and no project axiom, an elementary resolvent bound whose hypothesis asks for row sums *bounded* by $\alpha < 1$ and never constant: for a nonnegative matrix C over a finite index type, supported on pairs at distance at most one,

$$\sum_{n < N} (C^n)_{ij} \leq \frac{\alpha^{\text{dist}(i,j)}}{1 - \alpha} \quad \text{for every } N,$$

with no reference of any kind to the cardinality of the index type. We exhibit a non-vacuity witness and record the oracle.

Separately, and by measurement rather than proof, we remove the wall from where it was being placed. For the coupled kernel the second-eigenvalue ratio $\text{specRatio}(L)$ saturates strictly below one in each of six cells tested inside Onsager’s disordered region — with the spatial weight switched on — and reaches one to six decimal places in the two ordered cells tested. The control at zero spatial coupling reproduces the *proved* identity $\text{specRatio} = \tanh \beta$ to six decimals, so the harness is calibrated against a theorem rather than against itself. These data are eight cells at $L \leq 12$ with an extrapolation: they refute the attribution of the degeneracy to the spatial weight, and they do *not* locate the transition, which would need a dense sweep and finite-size scaling that we do not attempt.

What is not proved. The volume-uniform bound on specRatio for the coupled kernel is *not* established here. Dobrushin’s uniqueness theorem is classical [1, 3]; nothing in this paper is new mathematics. What is offered is a mechanised, non-vacuous ingredient, a measured relocation of a wall, and an exactly named remaining obligation. No consequence for Yang–Mills theory is stated, suggested, or implied.

1 What this paper claims, and what it does not

Claimed, machine-checked. Theorem 3.1 and its witness, in [YangMills/OS/DobrushinMatrix.lean](#). Every declaration is oracled and depends on exactly Lean’s three standard axioms.

Claimed, measured. The saturation table of Section 4. These are floating-point computations at finite extent with a documented extrapolation; they are evidence, not proof, and they are labelled as such wherever they appear.

Claimed, about the record. Section 2 states precisely what the source theorem `coupled_rowSums_not_constant` does and does not license. This is a statement about a formal development, and it is verifiable by reading that development.

Not claimed. That `specRatio` of the coupled kernel is bounded away from one uniformly in the extent. That the window derived from Dobrushin’s condition is optimal — it is not, and Section 6 says by how much. That any of this bears on the Yang–Mills mass gap.

2 The setting, and the theorem that was read too widely

Fix an extent L and consider configurations $\sigma \in \{0,1\}^L$ of one slice. The development under discussion carries a time-bond kernel that factorises over sites,

$$K_\beta(\sigma, \tau) = \prod_{j < L} b_\beta(\sigma_j, \tau_j), \quad b_\beta(s, s') = e^{\pm\beta},$$

and an arbitrary strictly positive slice weight w , the Gibbs weight of a space-time configuration being the product of the slice weights and the time bonds. When w is itself a product over spatial bonds with coupling γ , the system is the anisotropic two-dimensional Ising model.

Two facts from that development matter here.

Proposition 2.1 (proved elsewhere in the development). *At $\gamma = 0$ and for every $L \geq 1$, `specRatio` = $\tanh \beta$ exactly.*

The proof is Schur’s test [6]: the decoupled kernel has *constant* row sums, and the loss on the odd part of an observable is exactly compensated by the odd bond eigenvalue at every step.

Proposition 2.2 (`coupled_rowSums_not_constant`). *At $\gamma > 0$ the row sums of the coupled kernel are not constant: two explicit configurations of a two-site slice have different row sums.*

Proposition 2.2 is labelled THE OBSTRUCTION in the source. *It obstructs Schur’s test.* It says nothing whatever about `specRatio`, it is proved on a two-site witness, and no statement quantified over L appears in it.

A stronger reading — that the spatial weight destroys uniformity — was entered into the development and then *retracted of record* as unproved, leaving behind only a measurement of degeneracy.

What is documented, and can be checked. The retraction of the sentence from the manuscript is recorded at Addendum 561 of the development’s verification ledger; its removal from the module docstring, which the first retraction had missed, at Addendum 563; and its survival in a hand-typed submission form, caught only because the form was read in full before delivery, at Addendum 568. Three removals were required for one sentence, and the third was the one that would have been published.

What is inference, and is labelled as such. We observe that no statement quantified over L for the coupled kernel was attempted in the interval, and we attribute this to the retracted reading continuing to govern target selection. That attribution is ours; it is not established by a diff, and no artefact in the development records a decision not to attempt something. The verifiable half of the observation is the narrower one, and it is enough for this paper: the guards that caught the sentence three times compare identifiers and substrings inside files, so whatever else was carrying the reading was not a file.

3 The volume-free resolvent bound

Theorem 3.1 (machine-checked). *Let ι be a finite type, $d : \iota \times \iota \rightarrow \mathbb{N}$ satisfy $d(i, i) = 0$ and the triangle inequality, and let $C : \iota \times \iota \rightarrow \mathbb{R}$ satisfy*

- (i) $C_{ij} \geq 0$ for all i, j ;
- (ii) $C_{ij} = 0$ whenever $d(i, j) > 1$;
- (iii) $\sum_j C_{ij} \leq \alpha$ for all i , with $0 \leq \alpha < 1$.

Then for all i, j and every $N \in \mathbb{N}$,

$$\sum_{n \in [0, N)} (C^n)_{ij} \leq \frac{\alpha^{d(i, j)}}{1 - \alpha},$$

and consequently $\sum_n (C^n)_{ij} \leq \alpha^{d(i, j)} / (1 - \alpha)$.

Proof. Three steps, each an induction on n . Nonnegativity is preserved by powers. Row sums multiply: expanding (C^{n+1}) and exchanging the two finite sums, $\sum_j (C^{n+1})_{ij} = \sum_k (C^n)_{ik} \sum_j C_{kj} \leq \alpha \sum_k (C^n)_{ik} \leq \alpha^{n+1}$, so each entry, being at most its row sum, is at most α^n . Support: if $n < d(i, j)$ then $(C^n)_{ij} = 0$, since in $(C^{n+1})_{ij} = \sum_k (C^n)_{ik} C_{kj}$ each k has either $d(i, k) > n$, killing the first factor by induction, or $d(i, k) \leq n$, whence $d(i, j) \leq d(i, k) + d(k, j) \leq n + d(k, j)$ forces $d(k, j) > 1$ and kills the second. Combining, $(C^n)_{ij} \leq \alpha^n$ when $n \geq d(i, j)$ and vanishes otherwise; splitting $[0, N)$ at $d(i, j)$ and reindexing the surviving block by $n = d(i, j) + m$ gives $\alpha^{d(i, j)} \sum_{m < N - d(i, j)} \alpha^m$, and the geometric sum is at most $(1 - \alpha)^{-1}$. $\square$

Remark 3.2 (where the volume-freeness is). *The cardinality of ι appears nowhere in the hypotheses or the conclusion. That is the entire content of the theorem: the index type may be as large as one likes and neither α nor the exponent notices. Every constant is read off single-site data.*

Remark 3.3 (the hypothesis that matters here). *Hypothesis (iii) asks for row sums bounded by α . It never asks for them to be constant. Proposition 2.2 refutes constancy and leaves (iii) untouched: it is precisely the crack through which the coupled kernel passes after Schur's test has died.*

Proposition 3.4 (non-vacuity, machine-checked). *On the three-point chain with d the graph distance and $C_{ij} = 1/4$ for $d(i, j) \leq 1$ and 0 otherwise, hypotheses (i)–(iii) hold with $\alpha = 3/4 < 1$, the matrix is not the zero matrix, the two endpoints are at distance exactly two, and the conclusion reads: the whole resolvent series between the endpoints is at most $9/4$.*

The witness is not decoration. Hypotheses (ii) and (iii) are jointly satisfiable by the zero matrix, for which the conclusion is trivial, and by data with $d \equiv 0$, for which the exponent is trivial. Proposition 3.4 excludes both.

4 The measurement, and where the wall is

This section is numerical. Nothing in it is proved, and nothing in it is used by Section 3.

Let $T_L = W_\gamma^{1/2} K_\beta W_\gamma^{1/2}$ be the symmetrised coupled kernel on $\{\pm 1\}^L$ with free spatial boundary, and $r(L) = |\lambda_1|/|\lambda_0|$ its second-eigenvalue ratio, which is `specRatio` of the development. We compute $r(L)$ for $L = 2, \dots, 12$ by dense symmetric eigendecomposition and extrapolate by Aitken.

β	γ	$\sinh 2\beta \sinh 2\gamma$	$r(2)$	$r(12)$	$r(\infty)$
<i>controls at zero spatial coupling</i>					
0.20	0.00	0.0000	0.197375	0.197375	$\tanh \beta = 0.197375$
0.40	0.00	0.0000	0.379949	0.379949	$\tanh \beta = 0.379949$
<i>disordered</i>					
0.10	0.10	0.0405	0.1096	0.1209	0.1214
0.20	0.20	0.1687	0.2360	0.2895	0.2922
0.30	0.30	0.4053	0.3733	0.5127	0.5230
0.40	0.40	0.7887	0.5122	0.7813	0.8162
0.10	0.80	0.4783	0.1651	0.4013	0.4570
0.80	0.10	0.4783	0.7246	0.8031	0.8075
<i>ordered</i>					
0.60	0.60	2.2785	0.7489	0.9992	1.000003
0.80	0.80	5.6433	0.8893	0.999999	1.000000

$r(\infty)$ is an Aitken extrapolation of the last three extents. Values above one in the ordered rows are extrapolation overshoot at the sixth decimal and are reported as measured.

Each of the six cells tested inside Onsager’s disordered region, $\sinh 2\beta \sinh 2\gamma < 1$, saturates strictly below one, with increment ratios near 0.6; both ordered cells tested reach one to six decimal places. The control at $\gamma = 0$ reproduces $\tanh \beta$ to six decimals, which is Proposition 2.1 — the harness is calibrated against a theorem of the development rather than against itself.

What these data support, stated at their strength and no further. They are consistent with the degeneracy being a property of the *ordered* phase, and they are *not* consistent with the degeneracy being caused by switching on the spatial weight: six disordered cells with the weight switched on fail to degenerate. That second, negative reading is the one this section is for, and it is the one that reopened the question.

What they do not support. They do not locate the transition. Eight cells at $L \leq 12$, with a three-term Aitken extrapolation whose overshoot is visible in the ordered rows above, cannot place a phase boundary; doing so would require a dense sweep near $\sinh 2\beta \sinh 2\gamma = 1$, finite-size scaling, and control of the extrapolator, none of which is attempted here. A reader should take from this section that the wall is *not* where the development had been treating it as being, and nothing about where it is.

Gates. Three gates were registered in [scripts/judge_dobrushin.py](#) together with the lane charter, before a line of the Lean of Section 3 was written. Each exits non-zero on failure; none bundles a theorem with its witness; the bound gate predicts a number rather than sampling. One of them was later found not to exercise the hypothesis of Remark 3.3 — it normalised every row to *exactly* α — and was redesigned to carry unequal row sums, strictly harder, declared in its own docstring and committed before being run. The gates emit a verdict only after an

explicit counter confirms that the expected number of checks actually executed, because an empty failure list is also what zero checks produce. No acceptance decision anywhere in this lane depends on a Python `assert`.

5 What remains, named exactly

Theorem 3.1 is an engine, not a conclusion. Turning it into a volume-uniform statement about `specRatio` requires, in order:

1. the single-site Dobrushin coefficient of the model, giving the explicit window $2 \tanh \beta + 2 \tanh \gamma < 1$;
2. Dobrushin’s comparison estimate [2, 3] — exponential decay of connected correlations at a rate free of the extent. **This is the bottleneck**; everything else is assembly;
3. transport of that decay into the operator formulation.

Step 1, measured but not proved

A two-state site in field h has conditional $P(+1) = (1 + \tanh h)/2$, and flipping one neighbour across a bond of strength J moves that field to $h - 2J$. The claim of step 1 is the identity

$$\sup_{h \in \mathbb{R}} |P_h(+1) - P_{h-2J}(+1)| = \tanh |J|, \quad \text{attained at } h = J.$$

A gate predicting this *identity* — not a range, and not a threshold chosen afterwards — was committed before the quantity was computed even once. It checks the attained value against $\tanh |J|$, checks that a sweep of $2 \cdot 10^5 + 1$ fields never beats the attained value, and fails on any violation.

J	predicted $\tanh J $	attained at $h = J$	max violation over sweep
0.05	0.049958374958	0.049958374958	0.000e+00
0.20	0.197375320225	0.197375320225	0.000e+00
0.50	0.462117157260	0.462117157260	5.551e−17
1.00	0.761594155956	0.761594155956	0.000e+00
2.00	0.964027580076	0.964027580076	0.000e+00
−0.70	0.604367777117	0.604367777117	0.000e+00

The Lean proof of this identity was attempted and does not compile. The intended mechanism is $\tanh a - \tanh b = \sinh(a-b)/(\cosh a \cosh b)$ together with $\cosh a \cosh b \geq \cosh^2((a-b)/2)$, which minimises the denominator exactly at fields symmetric about zero; the attempt fails on library naming and on several tactic steps, and is recorded with its diagnosis rather than removed. Consequently step 1 carries the label *measured*, not *proved*, and no theorem of this paper depends on it.

Sharpness is the reason the gate checks attainment separately. A coefficient that were merely an upper bound would leave the window undetermined at its own boundary; the supremum being reached is what would make $\tanh |J|$ the constant rather than a number above it.

The consumer already exists in the development. It takes, over a family of finite volumes, decay at a *common* rate $r < 1$ and returns a *common* mass $m > 0$ with $\|\text{projected transfer}\| \leq$

e^{-m} for every member of the family. Its quantifier order is worth stating because it is load-bearing: the prefactor C may depend on the volume and on the observable, and only r is shared. The development’s own standing objection — that a bound $C(L)\rho^N$ with $C(L) \rightarrow \infty$ says nothing — therefore does not apply to it, the conclusion being an operator-norm bound.

A design decision is recorded here rather than discovered later: the vacuum pairing is an infinite-time object, so step 3 must proceed through finite time extent with explicit boundary control, taking the limit only at the level of the bound. No infinite-volume state is constructed anywhere in this paper, and the word *clustering* is accordingly not used for one.

6 Prior art, and the exact delta

Dobrushin’s uniqueness theorem is classical [1, 2, 3] and nothing in this paper is new mathematics. We locate the contribution precisely.

Not explicit constants. The cell here is the anisotropic two-dimensional Ising model, whose transfer spectrum has been available in closed form since Onsager [4] and Kaufman [5]. A constant obtained from Dobrushin’s condition is *worse* than the exact answer — on the cells we measured, by a factor between roughly two and thirty-six. Any claim to have improved a constant for this model would be false, and we make none.

Generality, instead. Dobrushin’s coefficient is read off the single-site conditionals of an *arbitrary* positive finite-range slice weight, which is what the development’s Gibbs weight quantifies over, and where the exact solution is silent.

Mechanisation. A search conducted on 1 August 2026 returned no formalisation of Dobrushin’s uniqueness condition in any proof assistant. The same search does return a Lean 4 formalisation of the transverse-field Ising chain [9], with momentum-mode decomposition and a Jordan–Wigner transformation; it concerns the quantum chain rather than the classical two-dimensional transfer matrix, and displaces nothing here, but it means a blanket claim that the Ising model has not been formalised would not survive review, and we make none. A formalisation of the Perron–Frobenius theorem for irreducible nonnegative matrices also exists [8], as does a mechanisation of the Osterwalder–Schrader axioms for the Gaussian free field [10]. Reflection positivity and finite-lattice spectra for related models have been treated without mechanisation [11].

7 Limitations

Theorem 3.1 is elementary and its proof is three inductions; its value is that it is machine-checked, non-vacuous, and has the hypothesis the dead method violated. Section 4 is numerics at $L \leq 12$ with an Aitken extrapolation: it locates a wall, it does not prove where the wall is, and an extrapolation can mislead near a transition. The window of Section 5 is not optimal and is never called sharp. Step 2 of Section 5 is open, and until it closes, the volume-uniform statement about specRatio for the coupled kernel remains unproved.

8 Reproducibility

Anchor. All sources cited above are read at commit 1e4f9c0f3e60d42c1929482ca98b00c140d2b038 of <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>. Toolchain leanprover/lean4:v4.29.0-rc6; Mathlib pinned at 07642720480157414db592fa85b626dafb71355b. A

runner verifies all three of these before compiling anything and aborts on any mismatch, so that a stale cache cannot masquerade as a reproduction.

Two planes, and what each measured. The development machine measured the core at 8465 jobs with the module’s import removed and 8466 with it restored — the delta obtained by rebuilding the same tree twice, not by subtracting against a baseline measured elsewhere. A fresh Linux clone at the anchor then reproduced 8466 jobs independently, with zero errors.

Oracle. On the development machine, the fourteen declarations of the module were checked individually and each depends on exactly `[propext, Classical.choice, Quot.sound]`, with no `sorryAx` and no project axiom. On the Linux plane the repository-wide oracle ran to completion with exit code zero over 2794 axiom reports and zero occurrences of `sorryAx`.

Independent hash agreement. A second desk, working from its own check-out on a different operating system and without coordination on the value, reported `12c1fcdb12e39e46381fa372a102503f5d63522ecbae27c0b3911671a7c04eb7` at 11467 bytes for the module. The Linux run at the anchor computed the same digest and the same size. The oleans, the axiom reports and the elaboration therefore refer to the same bytes, which is the only reason any of the numbers above can be compared at all.

Certifiers. Both gate scripts were run in `normal` and in `optimized (python -O)` mode on the Linux plane. All four runs exited zero: `judge_dobrushin.py` reported 29 of 29 checks in each mode, and `judge_dobrushin_d2.py` reported 18 of 18. No acceptance decision in either script depends on a Python `assert`; each counts the checks it performed and refuses to print a verdict when fewer ran than expected, because an empty failure list is also what zero checks produce.

Child processes. Every child of the runner writes a sentinel holding exactly one line, its real decimal exit code, captured after termination and written atomically with validation. Four states are distinguished on read: `absent`, `malformed`, `non-zero`, `zero`. A run in which every sentinel reads `zero` is reported as a *pass candidate*, never as a *pass*, because the logs still have to be read.

One module in the anchor does not compile. `DobrushinCoefficient.lean`, the attempt at step 1 of Section 5, exits non-zero. It is imported neither by the core nor by the oracle, so the measurements above are unaffected — and it is retained with its diagnosis rather than deleted, this development’s standing rule being that a measured failure is committed and not removed. No claim of this paper rests on it.

References

- [1] R. L. Dobrushin, *The description of a random field by means of conditional probabilities and conditions of its regularity*, Theory of Probability and its Applications **13** (1968), 197–224. The uniqueness condition and the comparison estimate whose mechanised form is step 2 of Section 5. The theorem of this paper is classical and due here.
- [2] R. L. Dobrushin, *Prescribing a system of random variables by conditional distributions*, Theory of Probability and its Applications **15** (1970), 458–486.
- [3] B. Simon, *The Statistical Mechanics of Lattice Gases, Volume I*, Princeton University Press, 1993. Chapter V gives the Dobrushin uniqueness condition, the interdependence matrix, and the exponential decay of correlations it yields; the single-bond coefficient $\tanh |J|$ for the Ising cell and the resulting window $2 \tanh \beta + 2 \tanh \gamma < 1$ are the standard specialisation.

- [4] L. Onsager, *Crystal statistics I: a two-dimensional model with an order-disorder transition*, Physical Review **65** (1944), 117–149. The exact solution, and the source of the line $\sinh 2\beta \sinh 2\gamma = 1$ used as the reference boundary in Section 4.
- [5] B. Kaufman, *Crystal statistics II: partition function evaluated by spinor analysis*, Physical Review **76** (1949), 1232–1243. The transfer-matrix spectrum in closed form. This is the reason Section 6 claims generality rather than constants: for this cell the exact second eigenvalue is available and the Dobrushin constant is worse than it.
- [6] I. Schur, *Bemerkungen zur Theorie der beschränkten Bilinearformen mit unendlich vielen Veränderlichen*, Journal für die reine und angewandte Mathematik **140** (1911), 1–28. The test whose constant-row-sum form is the method discussed in Section 2.
- [7] E. Seneta, *Non-negative Matrices and Markov Chains*, revised printing, Springer, 2006. Standard reference for nonnegative matrices, ergodicity coefficients and the resolvent series $\sum_n C^n = (I - C)^{-1}$ of which Theorem 3.1 is a locally-supported, entrywise refinement.
- [8] The mathlib Community, *The Lean mathematical library*, Proceedings of CPP 2020, 367–381. The library against which the development is pinned; it carries a formalisation of the Perron–Frobenius theorem for irreducible nonnegative matrices.
- [9] *A Machine-Verified Proof of a Quantum-Optimization Conjecture*, arXiv:2606.29687 (2026). A Lean 4 formalisation of the Ising chain with momentum-mode decomposition and the Jordan–Wigner transformation. It concerns the transverse-field quantum chain rather than the classical two-dimensional transfer matrix, and displaces nothing here; it is cited because a blanket claim that the Ising model has not been formalised would be false.
- [10] *Formalization of Quantum Field Theory*, arXiv:2603.15770 (2026). Osterwalder–Schrader axioms for the Gaussian free field in Lean 4; the nearest mechanised neighbour on the constructive side, and free rather than interacting.
- [11] *Spectrum and reflection positivity on a finite lattice*, arXiv:2606.07922 (2026). Finite-lattice spectral and reflection-positivity results for a related model, not mechanised.